Pickled, fermented and brined vegetables: HMTc Standard

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CORRESPONDENCE

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PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for pickled, fermented and brined vegetables in Fruits, Vegetables, and Produce, evaluated on drained product as sold, wet weight, brine and packing liquid excluded from the analytical portion. 6 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only; the strictest applicable sovereign maximum level for the row's product form in the row's native basis, a disclosed read-across where the exact cell is unregulated, a substantiated not-material determination, or a control; never occurrence-derived)

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	100	government maximum level
Cadmium (Cd)	Tier 1	20	government maximum level
Total arsenic (tAs)	Tier 2	100	government maximum level

Inorganic arsenic (iAs)	Tier 1	60	human-grade read-across (Israel)
Mercury, total (tHg)	Tier 1	10	human-grade read-across (Hong Kong)
Methylmercury (MeHg)	Tier 1	control	measured every lot; no human-food maximum
Nickel (Ni)	Tier 2	400	human-grade read-across (European Union)
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	500	screening ceiling (Hong Kong)
Hexavalent chromium (Cr-VI)	Tier 2	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	100	government maximum level
Cadmium (Cd)	20	government maximum level
Total arsenic (tAs)	100	government maximum level
Inorganic arsenic (iAs)	60	Israel — human-food analog
Mercury, total (tHg)	10	Hong Kong — human-food analog

Methylmercury (MeHg)	control	No credible sovereign government sets a methylmercury maximum level for this product form, and no not-material determination is supportable for it: the routed corpus quantifies methylmercury in edible plant and fungal tissue (wang2021-houttuynia--mercury-china measured methylmercury in the leaf and root tissue of an edible leafy vegetable at an LOD of 0.002 ug/kg; stasinos2014-heavy--metals-tubers-review in root and tuber tissue; asensio2023--madrid-mushrooms-mercury), so the species is not absent from produce and the biological premise of a not-material determination is rebutted inside this very category. Every fresh vegetable class this row derives from (leafy, non-root and root/tuber) carries quantified methylmercury in the routed corpus and is held under the same control, and no evidence supports asserting that anaerobic brine fermentation is a methylation-free compartment; control is the fail-safe treatment where the affirmative claim is unsupported. Methylmercury is therefore held under mandatory analytical control rather than recorded not-material, and it remains bounded by identity below the published total-mercury ceiling for this row (read-across from Hong Kong Cap. 132V item 12, 10 ug/kg); methylmercury cannot exceed total mercury. No methylmercury ceiling is invented.
Nickel (Ni)	400	European Union — human-food analog

Tin (Sn)	control	Tin in this row is a packaging-migration analyte rather than a property of the food. Every sovereign tin maximum level in the register is conditioned on a canned (tinplate) presentation, and this row contains presentations at both extremes: table olives and sauerkraut are still widely packed in hermetically sealed tinplate, while pickles, kimchi and giardiniera are predominantly glass-jarred. Because the row's presentation determines whether any sovereign tin limit applies at all, HMTc publishes no standalone tin number for the row and instead makes the screen mandatory with an escalation to the canned-foods maximum level whenever the certified presentation is tinplate.
Aluminium (Al)	control	No credible sovereign government sets an aluminium contaminant maximum level for any fruit, vegetable or produce form in any jurisdiction. The register's aluminium limits are drinking-water and bottled-water parameters, infant-formula maximum levels, a food-contact plastics specific migration limit, toy migration limits, and permitted-additive levels for baked and starch foods; none of them regulates aluminium as a contaminant of produce. Aluminium is also the third most abundant element in the earth's crust and appears in produce largely as soil-derived and root-associated mineral content rather than as process contamination, so no near-neighbour read-across is supportable either. Under the government-limit-only method HMTc publishes no standalone aluminium number for this row.
Total chromium (Cr)	500	government maximum level

Hexavalent chromium (Cr-VI)	n/m	No credible sovereign government sets a hexavalent-chromium maximum level for any solid food in any jurisdiction. The register's only INGESTED Cr(VI) parameters are drinking-water maxima (Australia NHMRC 2025, 50 ug/L, nhmrc2025_crvi_drinking_water_-50; California SWRCB, 10 ug/L, us_ca_swrcb_2024_cr-vi_-chromium-hexavalent_mcl_10ugl and ca_swrcb_crvi_drinking_water_-10), which are water-basis parameters expressed per litre of a consumed liquid and supply no applicable ceiling for a solid produce row in this row's native basis; every remaining Cr(VI) row in the register is an article or material limit (toys, textiles, leather, fertilisers, electrical equipment). Chromium in produce is dominated by nutritionally essential Cr(III), and Cr(VI) is not stable in the plant matrix: fruit and vegetable tissue is an aqueous, organic-acid and ascorbate-rich reducing medium in which Cr(VI) is reduced to Cr(III), so a reported Cr(VI) value in this matrix reflects extraction conditions rather than the chromium species the consumer ingests. Hexavalent chromium is therefore bounded by the published total-chromium screen for this row (read-across from Hong Kong Cap. 132V item 7, 500 ug/kg) and controlled by reflex speciation rather than by a published number.

4 | Cite this standard

Pendergrass, K. "Pickled, fermented and brined vegetables." In Heavy Metal Tested & Certified (HMTc) Fruits, Vegetables, and Produce standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

References

[1] European Union — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Hong Kong — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Israel — human-food maximum levels
Binds iAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/pickled--fermented-vegetables. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-pickled-fermented-vegetables.